TECHNICAL REPORT

Experimental Reproducibility and Task-Specific Evaluation

WH-DRCS Hierarchical Sensing Framework

Supporting report for "Design and Optimization of a Doppler-Resilient ISAC Framework for UAV Hierarchical Sensing"

Prepared by Chengjuan Wan and coauthors

University of Chinese Academy of Sciences, Beijing, China

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REPORT STATUS Methodology, parameter settings, and reporting specifications are finalized. Held-out numerical result fields remain reserved for insertion after completion of the frozen test evaluation.

Executive Summary

This report provides the reproducibility record and task-specific evaluation protocol for the four-level WH-DRCS hierarchical sensing framework. It documents the data partition, parameter-selection policy, frozen thresholds and weights, input/output conventions, evaluation equations, and the artifact trail required to audit every reported result.

The protocol separates coefficient fitting, hyperparameter selection, and final testing at the trajectory level. The former aggregate “algorithmic accuracy” is replaced by metrics aligned with each task: an ROC curve and AUC for fundamental detection; confusion matrices and classification measures for target and configuration recognition; and rotor-speed and attitude MAE/RMSE with coverage for fine-grained estimation. The report also clarifies that the L3 profile, rather than L4, has the largest maximum unambiguous velocity; L4 instead prioritizes ambiguity suppression and estimation fidelity.

Table 1. Report control summary.

Report item	Specification
Evaluation scope	L1 detection, L2 classification, L3 configuration recognition, and L4 rotor-speed/attitude estimation
Dataset split	360 training, 120 validation, and 120 held-out test trajectories
Random seed	2026
Test-time retuning	None
Equal-resource waveform budget	16,384 training symbols
Numerical result status	Pending completion of the frozen held-out evaluation

REPORT STATUS This document intentionally distinguishes finalized methodology from pending numerical results. No value should be inferred from a blank or Pending field.

Contents

Purpose and Scope

This report supplements the implementation accompanying the manuscript:

> **Design and Optimization of a Doppler-Resilient ISAC Framework for UAV Hierarchical Sensing**

It specifies the complete evaluation protocol for the four hierarchical sensing tasks, including:

- trajectory-level dataset partitioning;
- parameter fitting and validation selection;
- frozen thresholds, weights, and DBSCAN settings;
- executable input/output conventions;
- task-specific performance metrics;
- treatment of rejected and misrouted samples;
- the interpretation of the L_3 and L_4 maximum unambiguous velocities;
- the files required to audit every reported result.

The protocol separates three different activities that must not share data:

Table 2. Data-partition roles and permitted operations.

Activity	Partition	Permitted operations
Parameter fitting	Training	Estimate weights and calibration statistics
Model selection	Validation	Select thresholds and fixed hyperparameters
Final reporting	Test	Evaluate the already frozen pipeline only

> **Methodological safeguard.** Module-level synthetic examples are used only to verify software execution. They are never mixed with the held-out test set and are never reported as experimental performance.

Evaluation Design

Dataset composition

The system-level evaluation combines:

- **600 SUG-UAV trajectory realizations**, providing UAV body trajectories and motion labels;
- the **NIST Quasi-Deterministic (QD) channel model**, providing physically based propagation characteristics;
- the **NIST Integrated Sensing and Communication Physical Layer Model (ISAC-PLM)**, generating the received physical-layer observations;
- **93 dynamic scattering centers** for rotor and body micro-motion modeling;
- an SNR evaluation range of **20–40 dB**.

The same trajectory, channel realization, noise seed, and ground truth are reused when comparing different waveforms or hierarchical configurations. Equal-resource waveform comparisons use a common budget of $128 \times 128 = 16{,}384$ training symbols.

Trajectory-level partition

The split is performed before frame generation, window extraction, or augmentation. Consequently, no segment derived from one trajectory can appear in more than one partition.

Table 3. Trajectory-level dataset partition.

Partition	Number of trajectories	Ratio	Function
Training	360	60%	Fit learned coefficients and estimate calibration statistics
Validation	120	20%	Select thresholds and non-learned hyperparameters
Test	120	20%	Produce final task-specific results

Table 4. Frozen data-split settings.

Split property	Frozen setting
Random seed	2026
Split unit	Complete trajectory
Stratification	Available target, airframe, and motion-state labels
Test-set access during development	Prohibited

The workflow is deliberately sequential: coefficients are fitted on the training partition, thresholds are selected on the validation partition, and the complete backend is frozen before the test partition is evaluated.

Evaluation modes

Two complementary modes are retained so that early-stage errors are not hidden:

1. **End-to-end evaluation:** every eligible test trajectory enters at L_1. An incorrect rejection or routing decision is counted as a downstream failure.
2. **Stage-conditional evaluation:** a stage is evaluated only on samples with the required valid input or ground-truth upstream label. This measures the intrinsic performance of that stage.

The end-to-end result is the primary system metric. Stage-conditional results may be reported as diagnostic evidence but must be labeled explicitly.

Statistical reporting

- Confusion matrices are released as both raw counts and row-normalized percentages.
- Macro-averaged metrics give equal weight to every class.
- Error metrics retain per-trajectory values before aggregation.
- When confidence intervals are reported, use 1,000 trajectory-level bootstrap resamples with seed `2026` and the percentile 95% interval.
- Rejected estimates are never silently removed; their frequency is reported through coverage or rejection rate.

Software Environment

Core dependencies

Table 5. Software dependencies and their roles.

Component	Requirement	Role
Python	3.8 or later	Hierarchical sensing backend
NumPy	Recorded in requirements.txt or the run manifest	Numerical processing
SciPy	Recorded in requirements.txt or the run manifest	Signal processing and statistics
scikit-learn	Recorded in requirements.txt or the run manifest	DBSCAN and evaluation metrics
NIST ISAC-PLM	External dependency	Physical-layer observation generation

Component	Requirement	Role
NIST QD channel model	External dependency	Propagation and channel realization

Install the Python dependencies from the repository root:

```
python -m pip install numpy scipy scikit-learn
```

Before a formal evaluation, archive the exact environment:

```
python -VV > results/python_version.txt
python -m pip freeze > results/python_packages.txt
git rev-parse HEAD > results/git_commit.txt
```

The ISAC-PLM and NIST QD software version or commit identifier must also be recorded in the run manifest because these upstream packages are distributed separately.

Data Interface

ISAC-PLM observation format

`ISACPLM_DataParser` reads a line-delimited JSON file. Each line represents one time-ordered observation:

```
{"axisVelocity": [12.4], "range": [98.7], "reflectionPower": [0.031]}
```

Table 6. ISAC-PLM observation fields.

Field	Required	Description
axisVelocity	Yes	Resolved axial-velocity observations in m/s
reflectionPower	Yes	Corresponding reflected-power observations
range	Recommended	Range observations in m; required for the complete L_1/L_2 feature set

Input order must follow slow time. When an observation contains multiple resolved scatterers, the current parser uses their mean value for the trajectory-level feature sequence.

Required ground truth

Table 7. Ground-truth requirements by sensing level.

Level	Ground-truth annotation
L_1	Target present / target absent
L_2	UAV / non-UAV

Level	Ground-truth annotation
L_3	Helicopter / quadrotor / hexarotor / octocopter
L_4	Rotor angular speeds and pitch/yaw/roll angles

Trajectory identifiers must remain attached to predictions and labels so that splitting, bootstrapping, and error aggregation are performed at the trajectory level.

Parameter Selection and Frozen Configuration

General policy

Table 8. Parameter-selection policy and test-time behavior.

Parameter category	Selection source	Behavior during testing
Protocol/frame parameters	Requested sensing level	Select one predefined profile
Learned coefficients	Training partition	Frozen
Decision thresholds	Validation partition	Frozen
Physical constraints	Platform or scenario definition	Frozen
Input-adaptive statistics	Deterministic analytical formula	Recomputed without fitting

No parameter is retuned for an individual test trajectory, SNR value, propagation scenario, UAV type, or comparison waveform. A new deployment may recalibrate parameters using a separate training/validation dataset, but its test set must remain untouched.

Hierarchical frame profiles

Table 9. Hierarchical frame profiles.

Level	Task	M	T_{PRI}	$v_{\max}$
L_1	Fundamental detection	8	$50.00\ \mu$s	25.0 m/s
L_2	Target classification	16	$25.00\ \mu$s	50.0 m/s
L_3	Configuration recognition	64	$10.00\ \mu$s	125.0 m/s
L_4	Attitude estimation	128	$15.00\ \mu$s	83.3 m/s

Shared upstream receiver settings are:

Table 10. Shared receiver settings.

Setting	Value
Doppler FFT window	Blackman–Harris
Symbol-sampling offset	0.75
CFAR threshold	6 dB

L_1: fundamental detection

The detector combines velocity smoothness S_v, range smoothness S_r, and reflection concentration $\mathcal{H}_p$:

$$\mathcal{C}_{L_1} = \mathbf{w}_1^T [S_v, S_r, \mathcal{H}_p]^T.$$

Table 11. Frozen parameters for L1 fundamental detection.

Quantity	Code symbol	Frozen value	Selection/derivation
Detection threshold	gamma_detect	0.85	Selected from the validation ROC
Acceleration bound	eta_a	15.0 m/s^2	Physical feasibility constraint
Smoothness scale	sigma_scale	2.0	Validation calibration
Metric covariance diagonal	_metric_covariance	[0.05, 0.02, 0.10]	Estimated on training data
IVW weights	$\mathbf{w}_1$	[0.25, 0.625, 0.125]	Normalized inverse covariance diagonal

The final target decision requires both

$$\mathcal{C}_{L_1} \geq \gamma_{\text{detect}} \quad \text{and} \quad a_t \leq \eta_a.$$

The threshold remains fixed for all reported SNRs and scenarios. During ROC generation, only the score threshold is swept; the acceleration constraint remains unchanged.

L_2: target classification

The classification feature vector is

$$\mathbf{x}_2 = [\mathcal{H}_A, \rho_{\text{micro}}, S'_r]^T,$$

and its linear discriminant is

$$\mathcal{C}_{L_2} = \mathbf{w}_2^T \mathbf{x}_2 + b_2.$$

Table 12. Frozen parameters for L2 target classification.

Quantity	Code symbol	Frozen value	Selection/derivation
Nyquist-boundary width	delta_v	2.0 m/s	Validation calibration
Linear weights	_w2	[0.45, 0.35, 0.20]	Fitted on the training partition

Quantity	Code symbol	Frozen value	Selection/derivation
Linear bias	_b2	-0.15	Fitted on the training partition
UAV threshold	gamma_uav	0.65	Selected using validation classification performance
Range-smoothness scale	sigma_scale	1.5	Validation calibration

$\mathcal{C}_{L_2}$ is a discriminant score, not a calibrated posterior probability. It is interpreted only through the frozen decision rule

$$\mathcal{C}_{L_2} \geq \gamma_{\mathrm{UAV}}.$$

L_3: configuration recognition

The strongest 10% of reflected-power samples are retained and clustered in the velocity domain. The adaptive DBSCAN neighborhood radius is

$$\epsilon = \beta\, \sigma_{\mathcal{V}} \sqrt[3]{\ln\left|\mathcal{V}_{\mathrm{peak}}\right|}.$$

Table 13. Frozen parameters for L3 configuration recognition.

Quantity	Code symbol	Frozen value	Selection/derivation
Retained-power percentile	power_percentile	90	Validation selection
Radius scale	epsilon_scale (β)	0.05	Validation selection
Minimum samples	min_samples	3	Validation selection
Distance metric	metric	Euclidean	Fixed definition
Kurtosis relaxation	tau_kurtosis	2.0	Validation selection
Confidence weights	_w3	[0.65, 0.35]	Validation selection
Recognition threshold	confidence_threshold	0.60	Validation selection

The value of ϵ is recalculated for every input because it depends on the measured spread and sample count. This is deterministic input adaptation, not test-time parameter training: β, `min_samples`, the percentile, metric, weights, and threshold remain fixed.

Table 14. Configuration labels and reference kurtosis values.

Cluster count	Configuration label	Reference kurtosis
2	Helicopter	1.5
4	Quadrotor	2.2
6	Hexarotor	2.8
8	Octocopter	3.5

L_4: rotor speed and attitude estimation

For each time-ordered rotor trace, local peak intervals $\{\Delta t_i\}$ provide the robust angular-speed estimate

$$\widehat{\omega}_k = \frac{2\pi}{\mathrm{median}(\{\Delta t_i\})}.$$

Table 15. Frozen parameters for L4 rotor-speed and attitude estimation.

Quantity	Code symbol	Frozen value	Selection/derivation
UAV arm length	arm_length_L	0.5 m	Simulated platform geometry
Minimum peak interval	min_peak_interval	5 ms	Validation selection
Peak prominence	peak_prominence	0.5	Validation selection
Reliability weights	_w4	[0.60, 0.40]	Validation selection
Reliability threshold	reliability_threshold	0.50	Validation selection
Vertical calibration coefficient	c_z	10^{-7}	Training calibration
Differential calibration coefficient	c_a	10^{-3}	Training calibration

In the unified pipeline, `dt` is inherited from the selected frame profile. A direct invocation of `L4_AttitudeEstimator` may override `dt` only when its rotor-resolved input traces have been resampled to a documented interval.

Selection objectives

The selected numerical values above are sufficient to reproduce inference. For model-selection reproducibility, the validation objectives are:

Table 16. Training and validation objectives by sensing stage.

Stage	Training/validation objective
L_1	Estimate calibration covariance on training data; select the operating threshold from the validation ROC
L_2	Fit the linear discriminant on training data; select the fixed decision threshold using validation macro-F1
L_3	Select clustering and confidence parameters using validation macro-F1; use lower rejection rate as the tie-breaker
L_4	Select peak and fusion parameters using validation rotor-speed and attitude error; select the reliability threshold from the validation error–coverage trade-off

Once selected, the complete parameter set is frozen before the test split is evaluated.

Reproduction Procedure

1. Install the backend

```
git clone https://github.com/wancj23/Unified-Hierarchical-Signal-Processing-for-UAV-ISAC.git
cd Unified-Hierarchical-Signal-Processing-for-UAV-ISAC
python -m pip install numpy scipy scikit-learn
```

2. Generate or prepare ISAC-PLM observations

Produce one line-delimited JSON file for each trajectory and preserve its trajectory identifier, waveform/configuration identifier, SNR, and ground-truth labels in the evaluation manifest.

3. Run an individual sensing level

```
python src/unified_hierarchical_processor.py \
  --input data/rda.json \
  --level L4 \
  --output results/l4_semantic_output.json
```

Change `--level` to `L1`, `L2`, `L3`, or `L4`. The processor executes all prerequisite stages and returns one structured JSON object.

4. Preserve predictions before aggregation

For every test trajectory, record:

- trajectory ID and ground truth;
- selected frame profile;
- all intermediate confidence or discriminant scores;
- every stage decision;
- the final class, rotor speeds, and attitude estimates where applicable;
- rejection status and reason.

Metrics must be recomputed from this preserved prediction file rather than copied manually from plots.

Task-Specific Evaluation Metrics

L_1: detection ROC

For threshold γ, define

$$P_D(\gamma) = \frac{TP(\gamma)}{TP(\gamma) + FN(\gamma)}, \qquad P_{FA}(\gamma) = \frac{FP(\gamma)}{FP(\gamma) + TN(\gamma)}.$$

Sweep γ over the observed $\mathcal{C}_{L_1}$ scores and report:

- the complete ROC curve (P_{FA}, P_D);
- AUC;
- the operating-point P_D and P_{FA} at `gamma_detect = 0.85`;
- positive and negative test counts;
- optional trajectory-bootstrap confidence intervals.

L_2: UAV/non-UAV classification

Report a *2 × 2* confusion matrix with **ground truth in rows** and **prediction in columns**:

Table 17. Definition of the L2 confusion matrix.

	Predicted non-UAV	Predicted UAV
True non-UAV	TN	FP
True UAV	FN	TP

Also report accuracy, precision, recall, specificity, and F1. The primary summary statistic is macro-F1, accompanied by the raw matrix so class imbalance remains visible.

L_3: configuration recognition

Report raw and row-normalized confusion matrices over the ordered class set:

```
Helicopter, Quadrotor, Hexarotor, Octocopter
```

The report includes:

- overall accuracy;
- macro-precision, macro-recall, and macro-F1;
- per-class recall;
- rejection rate;
- both end-to-end and stage-conditional results when both are available.

An upstream rejection or an L_3 clustering failure is counted as an incorrect end-to-end prediction rather than removed.

L_4: rotor-speed error

Because cluster identifiers are permutation-invariant, predicted rotors are first paired with ground-truth rotors using a minimum-cost one-to-one assignment based on absolute angular-speed error. Unmatched or rejected rotors are reported separately.

For trajectory j with K_j matched rotors,

$$\mathrm{MAE}_{\omega} = \frac{1}{\sum_j K_j} \sum_j \sum_{k=1}^{K_j} \left|\widehat{\omega}_{j,k} - \omega_{j,k}\right|,$$

$$\mathrm{RMSE}_{\omega} = \sqrt{\frac{1}{\sum_j K_j} \sum_j \sum_{k=1}^{K_j} \left(\widehat{\omega}_{j,k} - \omega_{j,k}\right)^2}.$$

Report both values in rad/s. A relative error may be included as a secondary metric, but it does not replace the absolute-unit errors.

L_4: attitude error

Let $q \in \{\text{pitch}, \text{yaw}, \text{roll}\}$. For J eligible test trajectories,

$$\mathrm{MAE}_{q} = \frac{1}{J} \sum_{j=1}^{J} \left|\hat{q}_j - q_j\right|,$$

$$\mathrm{RMSE}_{q} = \sqrt{\frac{1}{J} \sum_{j=1}^{J} \left(\hat{q}_j - q_j\right)^2}.$$

Errors are reported in degrees. Yaw error is wrapped to $[-180^{\circ}, 180^{\circ})$ before taking its magnitude; pitch and roll use direct signed differences within their defined ranges.

To prevent favorable selection bias, also report

$$\text{Coverage} = \frac{\text{number of accepted} L_4 \text{trajectories}}{\text{number of eligible} L_4 \text{trajectories}}.$$

Held-Out Test Results and Reporting Tables

Results status. Numerical fields in this section are intentionally marked *Pending*. They must be populated only with values generated from the frozen 120-trajectory test split; smoke-test outputs and the former aggregate “algorithmic accuracy” values are not valid substitutes.

L_1 result summary

Table 18. L1 held-out detection results.

Metric	Held-out value
Positive trajectories	*Pending*
Negative trajectories	*Pending*
ROC AUC	*Pending*
P_D at $\gamma_{\text{detect}} = 0.85$	*Pending*
P_{FA} at $\gamma_{\text{detect}} = 0.85$	*Pending*

L_2 result summary

Table 19. L2 held-out confusion matrix.

	Predicted non-UAV	Predicted UAV
True non-UAV	*Pending*	*Pending*
True UAV	*Pending*	*Pending*

Table 20. L2 held-out classification metrics.

Metric	Held-out value
Accuracy	*Pending*
Precision	*Pending*
Recall	*Pending*
F1	*Pending*

L_3 result summary

Table 21. L3 held-out configuration confusion matrix.

True \ Predicted	Helicopter	Quadrotor	Hexarotor	Octocopter
Helicopter	*Pending*	*Pending*	*Pending*	*Pending*
Quadrotor	*Pending*	*Pending*	*Pending*	*Pending*
Hexarotor	*Pending*	*Pending*	*Pending*	*Pending*
Octocopter	*Pending*	*Pending*	*Pending*	*Pending*

Table 22. L3 held-out configuration metrics.

Metric	Held-out value
Accuracy	*Pending*
Macro-precision	*Pending*
Macro-recall	*Pending*
Macro-F1	*Pending*
Rejection rate	*Pending*

L_4 result summary

Table 23. L4 rotor-speed and attitude error metrics.

Quantity	MAE	RMSE	Unit
Rotor angular speed	*Pending*	*Pending*	rad/s
Pitch	*Pending*	*Pending*	degree
Yaw	*Pending*	*Pending*	degree
Roll	*Pending*	*Pending*	degree

Table 24. L4 acceptance and coverage statistics.

Additional statistic	Held-out value
Eligible trajectories	*Pending*
Accepted trajectories	*Pending*
Coverage	*Pending*

Maximum Unambiguous Velocity Clarification

For the adopted monostatic Doppler model,

$$v_{\max} = \frac{\lambda}{4T_{\mathrm{PRI}}}.$$

At a 60 GHz carrier frequency, $\lambda \approx 5$ mm. The resulting values are:

Table 25. Maximum unambiguous velocity by configuration.

Configuration	T_{PRI}	$v_{\max}$	Interpretation
Standard Golay baseline	$14.15\ \mu$s	88.3 m/s	Baseline operating point
WH-DRCS L_3	$10.00\ \mu$s	125.0 m/s	Highest maximum unambiguous velocity
WH-DRCS L_4	$15.00\ \mu$s	83.3 m/s	Full-rank, high-fidelity sensing profile

The sensing hierarchy does **not** imply that every physical-layer metric increases monotonically from L_1 to L_4. In particular:

- L_3 uses the shortest PRI and therefore provides the largest $v_{\max}$;
- L_4 does not further extend $v_{\max}$ and is slightly below the standard Golay value;
- L_4 instead allocates the full $M = 128$ WH-DRCS set to strengthen ambiguity suppression, rotor separation, and fine-grained parameter estimation.

Thus, L_3 and L_4 represent different operating objectives. L_3 prioritizes the velocity extent needed for configuration recognition, whereas L_4 prioritizes signal fidelity for rotor-speed and attitude estimation. Any statement that L_4 "further extends" the maximum unambiguous velocity should be replaced by this trade-off interpretation.

Result Artifact and Audit Trail

The final repository should retain the following auditable structure:

```
results/
├── run_manifest.json
├── split_manifest.csv
├── python_version.txt
├── python_packages.txt
├── git_commit.txt
├── predictions/
│   └── held_out_predictions.csv
├── l1/
│   ├── roc_curve.csv
│   ├── metrics.json
│   └── roc_curve.png
├── l2/
│   ├── confusion_matrix_counts.csv
│   ├── confusion_matrix_normalized.csv
│   ├── metrics.json
│   └── confusion_matrix.png
├── l3/
│   ├── confusion_matrix_counts.csv
│   ├── confusion_matrix_normalized.csv
│   ├── per_class_metrics.csv
│   ├── metrics.json
│   └── confusion_matrix.png
└── l4/
    ├── rotor_speed_errors.csv
    ├── attitude_errors.csv
    ├── metrics.json
    └── estimation_error_summary.png
```

Minimum prediction-table columns

Table 26. Minimum schema for archived trajectory-level predictions.

Category	Required columns
Identity	trajectory_id, split, snr, scenario_id
Configuration	waveform, sensing_level, M, T_PRI, v_max
L_1	ground truth, $\mathcal{C}_{L_1}$, decision, rejection reason
L_2	ground truth, $\mathcal{C}_{L_2}$, predicted class
L_3	ground truth, K, ϵ, $\mathcal{C}_{L_3}$, predicted configuration
L_4	true/estimated rotor speeds, true/estimated angles, $\mathcal{C}_{L_4}$, acceptance flag

Reproduction Checklist

Before releasing or citing the final results, verify that:

- the split manifest contains exactly 360 training, 120 validation, and 120 test trajectories;
- no trajectory identifier appears in multiple partitions;
- all reported results use the same frozen parameter set;
- the test split was evaluated only after parameter selection ended;
- confusion matrices include every eligible sample and disclose rejections;
- rotor-speed matching is permutation-invariant;
- yaw errors use circular wrapping;
- every plotted value can be reconstructed from an archived CSV or JSON file;
- the environment and upstream simulator versions are recorded;
- all *Pending* fields in this document have been replaced with held-out results;
- the maximum-unambiguous-velocity discussion identifies L_3, not L_4, as the highest-$v_{\max}$ profile.

Citation

If this protocol or implementation supports your work, please cite the accompanying manuscript:

> C. Wan *et al.*, “Design and Optimization of a Doppler-Resilient ISAC Framework for UAV Hierarchical Sensing.”

Contact

Chengjuan Wan
University of Chinese Academy of Sciences, Beijing, China
wanchengjuan23@mails.ucas.ac.cn